1. COMPREHENSIVE OVERVIEW OF THE STATE OF THE PROBLEM AND EXISTING SOLUTIONS

Sepsis is a life-threatening clinical syndrome caused by a dysregulated host response to infection, leading to organ dysfunction, shock, and high mortality if not treated promptly. It remains one of the leading causes of death in intensive care units (ICUs), neonatal care units, and emergency departments worldwide. The burden is particularly significant in low- and middle-income countries, where delayed diagnosis, limited laboratory infrastructure, and high patient load further complicate management. In clinical practice, sepsis may progress rapidly from mild infection to severe systemic failure, making early detection and continuous monitoring essential for improving survival outcomes.

The timely diagnosis of sepsis is critically important because each hour of delay in appropriate treatment significantly increases mortality risk. Patients affected include neonates, elderly individuals, immunocompromised populations, surgical patients, and critically ill hospitalized individuals. Beyond patient outcomes, sepsis imposes substantial economic pressure on healthcare systems through prolonged ICU stays, repeated laboratory investigations, broad-spectrum antibiotic use, and increased resource utilization. If not identified at an early stage, sepsis can result in multi-organ failure, antimicrobial resistance due to empirical therapy, and long-term morbidity among survivors.

Current diagnostic approaches primarily rely on blood culture, molecular assays, and measurement of inflammatory biomarkers such as C-reactive protein (CRP), procalcitonin (PCT), interleukin-6 (IL-6), and lactate. Blood culture is considered the clinical gold standard for pathogen confirmation; however, it typically requires 24–72 hours and may yield false-negative results in low bacterial load or prior antibiotic exposure. Polymerase chain reaction (PCR)-based methods and multiplex molecular diagnostics offer faster pathogen detection with high sensitivity, but they require expensive instrumentation, trained personnel, and controlled laboratory settings. Conventional immunoassays such as ELISA and chemiluminescence analyzers are widely used for biomarker quantification, yet they involve multiple processing steps, reagent dependency, and centralized laboratory workflows. Recent research has also explored microfluidic biosensors, electrochemical sensors, and AI-assisted predictive models for early sepsis risk stratification.

Despite these advances, significant limitations remain. Most available systems are expensive, time-consuming, and infrastructure-dependent, restricting their use in emergency settings, rural hospitals, and point-of-care environments. Many methods focus on a single biomarker, whereas sepsis is a heterogeneous syndrome requiring multi-parameter assessment. Existing platforms may also suffer from sample preparation complexity, maintenance burden, delayed turnaround time, and reduced accessibility in resource-constrained settings. Furthermore, dependence on centralized diagnostics delays clinical decision-making during the critical early window of intervention.

These challenges establish a strong need for rapid, sensitive, portable, and low-volume diagnostic platforms capable of detecting multiple sepsis-related biomarkers directly from biological samples. Surface-Enhanced Raman Spectroscopy (SERS)-based sensing offers a promising alternative due to its high sensitivity, molecular fingerprint specificity, minimal sample preparation, and potential for miniaturized point-of-care deployment. By integrating nanostructured plasmonic substrates with biomarker analysis, the proposed work aims to develop a fast and label-free sensing strategy for sepsis biomarker detection. Such a system can reduce diagnostic delay, support timely treatment decisions, and improve accessibility of advanced diagnostics across diverse healthcare settings.

2. INTELLECTUAL PROPERTY (PATENT LANDSCAPING)

2.1 Overview of Existing Intellectual Property

The field of sepsis diagnostics has attracted significant intellectual property activity due to the urgent need for rapid and accurate detection technologies. Existing patents broadly cover biomarker assays, molecular diagnostics, microfluidic platforms, electrochemical biosensors, and optical sensing systems. Major industrial stakeholders include Roche, Abbott Laboratories, Siemens Healthineers, bioMérieux, and Thermo Fisher Scientific, while universities and research institutes have focused on biosensor and nanotechnology-enabled solutions. Patent activity related to Surface-Enhanced Raman Spectroscopy (SERS) is particularly concentrated in plasmonic substrates, signal enhancement nanostructures, label-based immunoassays, and portable Raman instrumentation.

2.2 Patent Landscape Analysis

A review of published patents indicates that several innovations are already protected. For example, patents on automated sepsis biomarker analyzers cover immunoassay systems for procalcitonin (PCT), C-reactive protein (CRP), and interleukin measurements. Molecular diagnostic patents protect multiplex PCR cartridges and nucleic-acid amplification workflows for bloodstream infection detection. In the optical sensing domain, patents such as **US 8,802,356** (nanostructured SERS substrate fabrication) and **US 10,113,180** (portable Raman detection systems) demonstrate protection around plasmonic nanostructures and compact Raman instrumentation. Other filings describe gold/silver nanoparticle functionalization, antibody-linked SERS tags, and machine-learning assisted spectral classification. These patents indicate that substrate engineering, assay chemistry, and integrated hardware platforms are key protected areas.

2.3 Commercial Products / Market Solutions

Commercially available systems for sepsis-related biomarker testing include Elecsys BRAHMS PCT, ARCHITECT PCT, VIDAS B·R·A·H·M·S PCT, and multiplex molecular platforms such as FilmArray Blood Culture Identification Panel. Raman hardware vendors such as B&W Tek and Thermo Fisher Scientific offer portable spectrometers that can be adapted for biosensing research. However, most marketed products remain centralized laboratory systems rather than low-cost bedside screening tools.

2.4 Limitations and Gaps in Existing IP

Although current patents and products provide strong technical foundations, limitations remain. Many solutions depend on expensive reagents, labeled assays, and proprietary consumables. Some systems are bulky, require trained personnel, and are unsuitable for decentralized settings. Existing SERS patents often emphasize substrate fabrication but not complete clinical workflows for multi-biomarker sepsis screening. In addition, robust AI-enabled interpretation, disposable low-cost chips, and direct plasma analysis with minimal preprocessing remain underdeveloped.

2.5 Opportunity for Innovation

Substantial innovation opportunities still exist in developing a **label-free, multi-biomarker, portable SERS platform** for rapid sepsis screening. Novelty may arise from optimized Ag/Au bimetallic substrates, biomarker panels tailored for early sepsis detection, machine-learning assisted spectral classification, smartphone-linked readout systems, and low-volume sample handling. A clinically oriented point-of-care workflow focused on affordability and rapid turnaround can provide differentiation without duplicating existing protected technologies.

2.6 Comparative Table

Patent / Product	Key Feature	Limitation	Relevance to Proposed Work
US 8,802,356	Nanostructured SERS substrate fabrication	Focus on substrate only	Useful for plasmonic design concepts
US 10,113,180	Portable Raman detection system	Generic sensing platform	Relevant for compact instrumentation
Elecsys BRAHMS PCT	Automated PCT quantification	Expensive centralized analyzer	Benchmark biomarker assay
FilmArray BCID Panel	Rapid pathogen ID from blood culture	High cartridge cost	Shows demand for rapid diagnostics
Proposed Work	Label-free SERS biomarker panel	Under development	Point-of-care sepsis screening

3. OBJECTIVES & METHODOLOGIES

3.1 OBJECTIVES

1. **To design and fabricate** a plasmonic Surface-Enhanced Raman Spectroscopy (SERS) substrate based on Ag/Au nanostructures for sensitive detection of sepsis-associated biomarkers.
2. **To develop and optimize** an experimental protocol for rapid Raman/SERS analysis of clinically relevant biomarkers such as C-reactive protein (CRP), procalcitonin (PCT), interleukin-6 (IL-6), tumor necrosis factor-alpha (TNF-α), lipopolysaccharide (LPS), lipoteichoic acid (LTA), and peptidoglycan-related markers.
3. **To analyze and validate** acquired Raman spectra using preprocessing, peak identification, and statistical or machine learning methods for biomarker discrimination and classification.
4. **To compare and evaluate** the proposed SERS-based sensing platform with conventional diagnostic methods in terms of sensitivity, speed, cost, portability, and point-of-care applicability.

3.2 METHODOLOGY

The project was initiated by identifying the clinical challenge of delayed sepsis diagnosis, particularly in critical care and neonatal settings where rapid intervention is essential. A detailed literature review was conducted on sepsis biomarkers, Raman spectroscopy, Surface-Enhanced Raman Spectroscopy (SERS), nanostructured plasmonic substrates, and existing laboratory diagnostic systems. This review established the need for a rapid, label-free, and low-cost sensing platform capable of early biomarker detection.

Based on the identified gap, a sensing architecture was proposed consisting of a nanostructured SERS-active substrate, a Raman spectroscopic acquisition system, spectral analysis tools, and an optional machine-learning based decision layer. Silicon wafers coated with silver, gold, or Ag/Au bimetallic nanostructures were considered as the sensing substrate due to their strong localized surface plasmon resonance and signal enhancement capability.

Analytical-grade biomarker standards and bacterial cell wall markers such as NAM, LPS, and LTA were selected for proof-of-concept experiments. Known concentrations of analytes were prepared in suitable solvents, and microliter-scale droplets were deposited onto the SERS substrate followed by controlled drying. Blank substrate measurements and solvent controls were also recorded to eliminate background contributions.

Raman spectral acquisition was performed using a microscope-coupled Raman system with appropriate laser excitation (e.g., 785 nm). Instrument parameters such as laser power, integration time, objective lens, accumulations, and spectral range were systematically optimized to maximize signal-to-noise ratio while preventing sample degradation. For each sample, replicate spectra were collected from multiple substrate locations to evaluate repeatability and spot-to-spot variation.

The acquired spectra were processed using computational tools such as Python, Origin, MATLAB, or equivalent scientific software. Preprocessing steps included dark subtraction, baseline correction, smoothing, normalization, and peak extraction. Characteristic Raman bands corresponding to target biomarkers were identified through comparison with literature and theoretical references.

Where applicable, supervised machine-learning models such as Support Vector Machine (SVM), Random Forest, Principal Component Analysis (PCA)-based classifiers, or neural network approaches may be used for spectral classification between healthy/control and sepsis-associated samples. Feature selection methods can be applied to identify the most diagnostically relevant Raman bands.

Validation of the proposed platform was carried out through sensitivity studies, concentration-dependent analysis, reproducibility assessment, and comparison with known biomarker signatures. Performance metrics included limit of detection (LOD), signal enhancement factor, accuracy, precision, sensitivity, specificity, repeatability, and analysis time.

The overall workflow of the project can be summarized as:

Problem Identification → Literature Review → SERS Substrate Design → Sample Preparation → Raman Data Acquisition → Spectral Processing → Biomarker Identification → Validation → Comparative Evaluation → Final Prototype Framework

This methodology establishes a translational pathway toward rapid and point-of-care sepsis biomarker diagnostics using SERS technology.

4. DETAIL OF WORK DONE

4.1 System Design / Proposed Framework

The proposed system was conceptualized as a rapid biosensing platform for sepsis biomarker detection using Surface-Enhanced Raman Spectroscopy (SERS). The framework integrates a plasmonic sensing substrate, Raman spectral acquisition unit, data preprocessing pipeline, biomarker signature library, and decision-support analytics. The overall workflow begins with sample deposition of biomarker-containing fluid onto a nanostructured SERS substrate, followed by laser excitation and collection of Raman scattered light. The acquired spectra are then processed to remove background noise and extract characteristic peaks corresponding to target biomarkers.

The sensing architecture was designed to support both fundamental biomarker studies and future point-of-care implementation. The modular structure includes: (i) substrate preparation module, (ii) sample handling module, (iii) optical acquisition module, (iv) spectral processing module, and (v) classification/interpretation module. This design allows independent optimization of each stage while maintaining an integrated translational workflow.

4.2 Work Completed on N-Acetylmuramic Acid (NAM)

1. Computational Methodology for Raman Simulation of N-Acetylmuramic Acid

To understand the vibrational behaviour of N-acetylmuramic acid (NAM) and to support the interpretation of the experimental Raman spectrum, Density Functional Theory (DFT) calculations were performed. The theoretical simulation was carried out using the Gaussian 16 computational chemistry package. DFT was selected because it provides a reliable balance between computational cost and accuracy for predicting molecular geometry, vibrational frequencies, and Raman activities of medium-sized organic biomolecules.

The hybrid exchange-correlation functional B3LYP was employed for all calculations. This functional is widely used in vibrational spectroscopy studies because it provides accurate frequency predictions for organic compounds containing oxygen, nitrogen, and carbon-based functional groups. Since NAM contains hydroxyl groups, an amide linkage, ether bonds, and a carboxylic acid moiety, B3LYP is suitable for reproducing its molecular vibrations.

The basis set used in the present study was 6-31+G(d,p). This basis set includes polarization functions and diffuse functions, which are important for describing the electron distribution around heteroatoms such as oxygen and nitrogen. The inclusion of these functions improves the modelling of polar bonds, hydrogen bonding tendencies, and vibrational motions involving lone-pair electrons.

The calculation was performed in two sequential stages:

1. Geometry Optimization
   The molecular structure of NAM was optimized to obtain the lowest-energy stable conformation. Tight convergence criteria were applied to ensure accurate bond lengths, bond angles, and torsional geometry.

2. Frequency and Raman Activity Calculation
   After geometry optimization, harmonic vibrational frequency analysis was performed. This step generated the theoretical vibrational frequencies and Raman activities corresponding to the normal modes of the molecule.

The molecular connectivity was explicitly defined in the input structure to preserve the correct bonding arrangement of the sugar ring, lactyl side chain, acetamide group, and hydroxyl substituents present in NAM.

The calculations were executed using 16 processor cores and 56 GB RAM to enable efficient convergence and frequency computation.

Simulation Parameters Used

Parameter	Value
Software	Gaussian 16
Method	DFT
Functional	B3LYP
Basis Set	6-31+G(d,p)
Geometry Optimization	Tight
Frequency Calculation	Raman
Resources	16 cores, 56 GB RAM
Molecule	N-Acetylmuramic Acid

Purpose of Simulation

The primary objective of the simulation was to generate a theoretical Raman spectrum of NAM and compare it with the experimentally measured powder spectrum. This comparison enables peak assignment, functional group identification, and validation of the molecular fingerprint of NAM.

Expected Outputs from Simulation

The Gaussian calculation produced the following outputs:

- Optimized molecular geometry
- Vibrational frequencies (cm^{-1})
- Raman scattering activities
- Normal mode information
- Theoretical stick spectrum for Raman analysis

These outputs were later processed and converted into a continuous simulated Raman spectrum for direct comparison with experimental data.

Molecular Structure of N-Acetylmuramic Acid and Components Considered for Simulation

2. Molecular Structure and Functional Components of NAM

N-acetylmuramic acid (NAM) is an amino sugar derivative and an essential structural unit of bacterial peptidoglycan. It forms part of the carbohydrate backbone of the bacterial cell wall and is covalently linked to peptide chains in many microorganisms. Because of its biological importance, NAM was selected as the target molecule for Raman and DFT-based vibrational analysis.

The molecular structure of NAM consists of a six-membered pyranose sugar ring substituted with several functional groups. These structural features are responsible for its characteristic Raman vibrations. During simulation, the complete molecular structure of NAM was considered so that all relevant vibrational modes arising from ring motion, side-chain motion, and functional group stretching or bending could be captured.

Major Structural Components Present in NAM

The following molecular units are present in N-acetylmuramic acid:

1. Pyranose Ring
   The central six-membered sugar ring forms the main carbohydrate framework of the molecule. This ring contributes significantly to low-frequency skeletal vibrations, ring deformation modes, and breathing motions.
2. Hydroxyl Groups (–OH)
   Multiple hydroxyl groups are attached to the sugar ring. These groups influence C–O stretching vibrations, O–H bending modes, and intermolecular hydrogen bonding behaviour.
3. Ether / Ring Oxygen
   The pyranose ring contains an oxygen atom as part of the cyclic ether structure. Vibrations involving the C–O–C linkage contribute strongly in the carbohydrate fingerprint region.
4. N-Acetyl Group
   NAM contains an acetamide substituent composed of a carbonyl group (C=O), methyl group (CH_3), and amide linkage (–NHCO–). This region is responsible for amide-related Raman bands and carbonyl stretching vibrations.
5. Lactyl Side Chain
   A lactyl substituent differentiates NAM from N-acetylglucosamine. This side chain contains additional oxygenated carbon atoms and contributes to ether and C–O vibrational modes.
6. Carboxylic Acid Group (–COOH)
   The molecule contains a terminal carboxylic acid group that gives rise to strong carbonyl stretching vibrations in the higher wavenumber region.

Molecular Formula and Structural Relevance

Property	**Value**
Compound Name	N-Acetylmuramic Acid
Abbreviation	NAM
Molecular Class	Amino Sugar
Biological Role	Peptidoglycan Component
Key Functional Groups	OH, C=O, Amide, Ether, COOH, CH_3

Why Full Molecule Simulation Was Necessary

The Raman spectrum of NAM is not generated by a single bond vibration. Instead, it arises from many coupled normal modes involving the entire molecular framework. Therefore, the complete molecule was simulated rather than isolated fragments. This allows realistic prediction of:

- Ring deformation modes
- Coupled C–O and C–C stretching
- Amide vibrations
- Carbonyl stretching
- CH bending and rocking
- Glycosidic-type ether vibrations
- Skeletal lattice motions

Structural Basis for Raman Peaks

Different parts of the molecule contribute to different spectral regions:

Spectral Region (cm^{-1})	Dominant Structural Contribution
200–600	Ring deformation and skeletal modes
600–900	Ring torsion, CH wagging, ether modes
900–1200	C–O, C–O–C, carbohydrate fingerprint
1200–1500	CH bending, amide III, side-chain modes
1500–1800	Amide II, carbonyl stretching

This structural understanding is essential for assigning the peaks observed in both the simulated and experimental Raman spectra.

Geometry Optimization and Frequency Calculation Procedure

3. Theoretical Simulation Workflow for NAM

After constructing the molecular structure of N-acetylmuramic acid, the next stage of the computational study involved geometry optimization followed by vibrational frequency analysis. These steps were necessary to obtain a physically meaningful structure and to calculate the Raman-active normal modes of the molecule.

The simulation workflow was carried out in a sequential manner so that the final theoretical spectrum accurately represented the optimized molecular configuration of NAM.

3.1 Geometry Optimization

The initial molecular coordinates of NAM were used as the starting structure for optimization. During this stage, the atomic positions were iteratively adjusted by the software until the total electronic energy reached a minimum. The optimized structure corresponds to a stable molecular conformation with energetically favorable bond lengths, bond angles, and torsional arrangements.

Tight optimization criteria were applied to improve structural accuracy. This ensures that the final geometry is suitable for subsequent vibrational analysis.

The purpose of geometry optimization was to:

- Remove unrealistic strain from the starting structure
- Obtain a stable minimum-energy conformation
- Refine bond distances and bond angles
- Generate a reliable structure for frequency calculations

The optimized geometry was then used as the input for Raman frequency analysis.

3.2 Vibrational Frequency Calculation

Once the optimized molecular structure was obtained, harmonic vibrational frequency calculations were performed. In this step, the second derivatives of the energy with respect to nuclear displacements were evaluated to generate the normal vibrational modes of the molecule.

Each normal mode corresponds to a specific collective atomic motion such as:

- stretching vibration
- bending vibration
- rocking motion
- torsional motion
- ring deformation
- coupled skeletal vibration

For every mode, the calculation generated:

- vibrational frequency (cm^{-1})
- Raman activity
- symmetry information
- atomic displacement pattern

Only positive frequencies indicate a stable optimized structure. The absence of imaginary frequencies confirmed that the geometry corresponded to a true local minimum on the potential energy surface.

3.3 Raman Activity Output

The frequency calculation also produced Raman scattering activities for each vibrational mode. These values describe how strongly each mode contributes to the Raman spectrum.

Modes with higher Raman activity generally produce stronger Raman peaks after spectral processing. These activities were later normalized and broadened to generate the final simulated Raman spectrum.

3.4 Computational Flow Summary

Step	Description
1	Build initial molecular structure of NAM
2	Perform geometry optimization
3	Confirm stable structure (no imaginary frequencies)
4	Run frequency calculation
5	Extract Raman activities
6	Generate theoretical Raman spectrum

Importance of This Step in the Study

Geometry optimization and frequency calculations form the foundation of the Raman simulation. Without an optimized structure, the predicted vibrational modes may not represent the real molecule accurately. By using the stable geometry of NAM, the calculated frequencies could be compared directly with experimental Raman peaks.

This comparison was later used to validate the molecular identity of NAM and assign functional groups to the observed Raman bands.

Output Files Generated

The Gaussian calculations typically generated the following files:

- input file (.gjf or .com)
- output file (.log or .out)
- checkpoint file (.chk)
- optimized geometry data
- frequency and Raman activity tables

These outputs were used in the post-processing stage to construct the final simulated spectrum.

Post-Processing of Simulated Data (Scaling, Broadening, Spectrum Generation)

4. Processing of Theoretical Raman Data

The raw output obtained from the DFT frequency calculation consists of discrete vibrational frequencies and their corresponding Raman activities. These values form a theoretical stick spectrum, where each line represents one normal mode of vibration. However, real experimental Raman spectra do not appear as sharp discrete lines. Instead, they contain broadened peaks due to instrumental resolution, molecular environment, and finite linewidth effects.

Therefore, post-processing of the theoretical data was performed to convert the raw computational output into a realistic simulated Raman spectrum that could be directly compared with the experimental powder spectrum.

4.1 Extraction of Frequency and Raman Activity Data

From the Gaussian output file, two important columns were extracted:

- Vibrational frequencies (cm^{-1})
- Raman activities

Only physically meaningful positive frequencies were retained. Any invalid or non-physical values were excluded from further analysis.

The extracted dataset served as the basis for spectral simulation.

4.2 Frequency Scaling Correction

DFT calculations using harmonic approximation generally overestimate vibrational frequencies. To correct this systematic deviation, a frequency scaling factor was applied.

The corrected frequency was calculated using:

$$\nu_{\text{scaled}} = \nu_{\text{calculated}} \times 0.96$$

where:

- $\nu_{\text{calculated}}$= raw DFT frequency
- ν_{scaled}= corrected theoretical frequency

The scaling factor of 0.96 was selected based on standard literature practice for B3LYP-based vibrational calculations and provided improved agreement with the experimental Raman data.

4.3 Selection of Spectral Range

The frequencies were filtered to the Raman region of interest used in the experimental analysis. Only peaks within the selected spectral window were retained for comparison.

Typical comparison range used in this work:

Spectral Window	**Purpose**
200–1800 cm^{-1}	Main molecular fingerprint region
200–3000 cm^{-1}	Extended full spectrum

The 200–1800 cm^{-1} region was used primarily for peak assignment and validation.

4.4 Normalization of Raman Activities

The Raman activities were normalized with respect to the maximum intensity so that all peaks could be compared on a relative scale.

The normalized intensity was calculated as:

$$I_{\text{norm}} = \frac{I}{I_{\text{max}}}$$

where:

- I = Raman activity of a mode
- I_{max} = highest Raman activity in the dataset

This normalization produces intensities ranging from 0 to 1.

4.5 Gaussian Broadening

Theoretical stick spectra were converted into continuous peaks using Gaussian broadening. This step simulates the natural linewidth of Raman bands and improves visual comparison with the measured experimental spectrum.

Each vibrational mode was represented by a Gaussian function centered at the scaled frequency.

A broadening parameter of:

$$\sigma = 10\ \text{cm}^{-1}$$

was used in this study.

This value provided a realistic peak width while preserving spectral resolution.

4.6 Generation of Final Simulated Spectrum

After scaling, normalization, and broadening, the final theoretical Raman spectrum was obtained as a continuous intensity curve across the selected Raman shift axis.

The processed simulated spectrum was then used for:

- overlay with experimental Raman spectrum

- peak matching analysis
- functional group assignment
- frequency error calculation
- validation of molecular fingerprint

Post-Processing Workflow Summary

Step	Operation
1	Extract frequencies and Raman activities
2	Remove invalid values
3	Apply scaling factor (0.96)
4	Select spectral range
5	Normalize intensities
6	Apply Gaussian broadening
7	Generate continuous Raman spectrum

Importance of This Stage

Without post-processing, direct comparison between theoretical and experimental spectra would be difficult because theoretical outputs are discrete and systematically shifted. By applying correction and broadening procedures, the simulated spectrum closely approximates the appearance of the real Raman spectrum.

This processed theoretical spectrum served as the main reference for validating the experimental NAM powder spectrum.

Experimental Raman Measurement of NAM Powder (Sample Preparation and Data Collection)

5. Experimental Raman Analysis of N-Acetylmuramic Acid Powder

To validate the theoretical Raman spectrum obtained from DFT calculations, experimental Raman measurements were performed using high-purity N-acetylmuramic acid powder. The experimental study was designed to establish the real molecular fingerprint of NAM and compare it with the simulated vibrational spectrum.

The use of a pure powder sample is important because it minimizes spectral interference from unrelated biomolecules and allows the intrinsic Raman bands of NAM to be observed.

5.1 Sample Material

The sample used in the present work was commercially obtained N-acetylmuramic acid powder with approximately 98% purity. The material was used as received for Raman measurements.

Property	Description
Compound	N-Acetylmuramic Acid
Form	Powder
Purity	~98%
Purpose	Experimental Raman Reference Sample

The high purity of the material ensured that the measured spectrum primarily represented NAM vibrations.

5.2 Sample Preparation

A small quantity of the powder was placed on a clean measurement surface suitable for Raman analysis. Care was taken to ensure a stable and uniform sample region for laser focusing.

No chemical treatment or additional reagents were required. The sample was analyzed in solid-state powder form.

General precautions followed during measurement:

- clean sample holder used
- stable powder region selected
- proper laser focus maintained
- repeated measurements performed for reproducibility

5.3 Raman Instrumentation

Raman spectra were collected using a laboratory Raman spectrometer equipped with a near-infrared excitation laser.

The instrument settings were varied systematically to optimize spectral quality.

Typical parameters used:

Parameter	Value
Excitation Source	785 nm laser
Measurement Type	Powder Raman
Spectral Range	200–3000 cm^{-1}
Variable Laser Power	5–80%

Parameter	Value
Variable Integration Time	5–25 s

The 785 nm excitation wavelength was suitable for reducing fluorescence interference while maintaining measurable Raman scattering intensity.

5.4 Acquisition Strategy

Multiple spectra were recorded under different combinations of laser power and integration time. This was done to determine the best acquisition conditions that provide:

- strong Raman signal
- low noise
- sharp peaks
- stable baseline
- no visible sample damage

A total set of spectra was collected across the parameter matrix, covering multiple power levels and acquisition times.

The filename convention stored these parameters directly in each dataset (for example: sec-25_power-80).

5.5 Objective of Experimental Measurement

The experimental Raman study had three main objectives:

1. Obtain the Raman fingerprint of pure NAM powder
2. Identify the optimal measurement parameters
3. Compare the measured peaks with DFT-simulated frequencies

Importance of Pure Powder Measurement

Unlike bacterial samples or complex mixtures, a pure NAM powder spectrum provides a direct molecular reference. This is valuable because bacterial spectra often contain overlapping contributions from proteins, lipids, nucleic acids, and other metabolites.

Therefore, the pure powder spectrum obtained in this study can serve as a control reference for future analysis of more complex biological samples.

Experimental Output

The Raman measurements generated raw spectral files containing:

- Raman shift axis (cm^{-1})
- detector intensity values
- dark-subtracted intensity data
- multiple spectra at different conditions

These raw files were later processed to select the highest-quality spectrum for final analysis.

Experimental Data Preprocessing and Best Spectrum Selection

6. Processing of Experimental Raman Spectra

The raw Raman spectra obtained from N-acetylmuramic acid powder measurements contained useful molecular information together with instrumental noise, background contribution, and intensity variations arising from different acquisition settings. Therefore, systematic preprocessing was performed before peak analysis and comparison with the simulated spectrum.

The objective of preprocessing was to improve signal quality, remove unwanted artifacts, and generate a clean normalized spectrum suitable for scientific interpretation.

6.1 Raw Data Organization

All measured spectra were stored as individual files corresponding to different combinations of laser power and integration time. Each file contained the Raman shift axis and measured intensity values.

The dataset included spectra recorded over a range of:

- integration times: 5 s to 25 s
- laser power: 5% to 80%

This enabled comparative evaluation of spectral quality under multiple conditions.

6.2 Dark Subtraction

The first preprocessing step involved the use of dark-subtracted intensity data. Dark subtraction removes detector background current and instrument noise present even in the absence of sample signal.

This improves baseline stability and enhances weak Raman features.

6.3 Spectral Smoothing

To reduce high-frequency random noise while preserving peak shape, Savitzky–Golay smoothing was applied. This method performs local polynomial fitting over a moving window and is widely used in Raman data processing.

Benefits of smoothing:

- improved signal clarity

- reduced noise fluctuations
- preserved peak positions
- maintained spectral shape

6.4 Baseline Correction

Raman spectra of powdered samples may contain baseline drift caused by fluorescence background, scattering effects, or detector response. To remove this slowly varying background, Asymmetric Least Squares (ALS) baseline correction was applied.

After baseline correction:

- peak intensities became more reliable
- weak peaks became clearer
- spectral comparison improved

6.5 Normalization

The corrected spectra were normalized relative to the maximum intensity. This step placed all spectra on a common intensity scale between 0 and 1, allowing direct comparison between different acquisition conditions.

Normalization was essential because absolute intensity depends on measurement settings such as laser power and exposure time.

6.6 Peak Detection

Automated peak detection was performed on the processed spectra to identify significant Raman bands. Peaks were selected based on prominence and spacing criteria to avoid false detection of noise.

The detected peaks were used for:

- spectral ranking
- peak assignment
- comparison with simulation
- creation of peak tables

6.7 Spectral Quality Metrics

To identify the best spectrum among all measured datasets, quantitative quality metrics were calculated.

The main metric used was Signal-to-Noise Ratio (SNR), which measures the strength of the Raman signal relative to background noise.

Additional factors considered:

- peak sharpness
- peak intensity
- full width at half maximum (FWHM)
- baseline quality

A composite score was then used to rank all spectra.

6.8 Best Spectrum Selection

Among all processed spectra, the highest-quality dataset was obtained at:

Parameter	Value
Integration Time	25 s
Laser Power	80%
Best File	sec-25_power-80_i-2.xlsx
Best SNR	95.91
Quality Interpretation	Excellent

This spectrum showed:

- strong signal intensity
- clear peak definition
- low noise
- stable baseline
- high reproducibility

Therefore, it was selected as the representative experimental Raman spectrum of NAM for final comparison with simulation.

Preprocessing Workflow Summary

Step	Operation
1	Load raw spectra
2	Use dark-subtracted signal

Step	Operation
3	Apply Savitzky–Golay smoothing
4	Perform ALS baseline correction
5	Normalize intensity
6	Detect peaks
7	Calculate SNR and quality metrics
8	Select best spectrum

Importance of This Step

Proper preprocessing is essential because theoretical validation and peak assignment depend on reliable experimental data. By selecting the highest-quality spectrum, the comparison with the DFT simulation became more accurate and scientifically meaningful.

The final processed experimental spectrum was then used for Raman peak assignment and validation of the NAM molecular fingerprint.

Experimental Raman Peaks, Functional Group Assignment, and Comparison with Simulation

7. Raman Peak Analysis and Validation of N-Acetylmuramic Acid

After preprocessing and selecting the highest-quality experimental spectrum, the Raman peaks of N-acetylmuramic acid were identified and interpreted. The final objective of this stage was to assign the major Raman bands to their corresponding molecular vibrations and compare them with the DFT-simulated frequencies.

The agreement between experimental and simulated peaks provides strong evidence that the measured powder spectrum corresponds to the expected molecular structure of NAM.

7.1 Major Experimental Raman Peaks Observed

The processed experimental spectrum of NAM exhibited multiple characteristic bands across the fingerprint region. The most relevant peaks selected for interpretation are listed below.

Peak Position (cm^{-1})	Relative Importance	Tentative Assignment
284.3	Strong	Ring deformation mode
458.2	Medium	Pyranose skeletal deformation
541.0	Strong	C–C–O bending / skeletal mode
772.7	Medium	CH wagging / ring vibration
830.2	Strong	Anomeric / glycosidic vibration

Peak Position (cm^{-1})	Relative Importance	Tentative Assignment
871.9	Very Strong	C–O–C stretching
955.9	Very Strong	Pyranose ring breathing
1024.0	Strong	C–O stretching
1155.4	Strong	C–O–C / carbohydrate vibration
1249.4	Medium	C–N / OH deformation
1345.9	Strong	CH deformation
1451.8	Strong	CH_2 / CH_3 bending
1547.9	Weak	Amide II / coupled vibration
1700.7	Medium	Carbonyl (C=O) stretching

These bands collectively form the Raman fingerprint of pure NAM.

7.2 Functional Group Interpretation

The Raman peaks arise from different structural components of the NAM molecule.

Carbohydrate Backbone Region (800–1200 cm^{-1})

This region is dominated by vibrations of the sugar ring, C–O bonds, and ether linkages.

Key peaks:

- 830.2 cm^{-1}
- 871.9 cm^{-1}
- 955.9 cm^{-1}
- 1024.0 cm^{-1}
- 1155.4 cm^{-1}

These peaks confirm the carbohydrate nature of NAM.

Ring and Skeletal Modes (200–700 cm^{-1})

This low-frequency region corresponds to collective motion of the pyranose ring and skeletal deformation.

Key peaks:

- 284.3 cm^{-1}
- 458.2 cm^{-1}

- 541.0 cm^{-1}

These modes support the presence of the cyclic sugar framework.

CH Deformation Region (1200–1500 cm^{-1})

This region is associated with methyl, methylene, and side-chain deformation vibrations.

Key peaks:

- 1345.9 cm^{-1}
- 1451.8 cm^{-1}

These bands are consistent with the acetyl and side-chain groups of NAM.

Carbonyl and Amide Region (1500–1800 cm^{-1})

This region corresponds to vibrations of the amide linkage and carboxylic acid carbonyl group.

Key peaks:

- 1547.9 cm^{-1}
- 1700.7 cm^{-1}

These peaks confirm the presence of the N-acetyl and acid functionalities.

7.3 Comparison with Simulated Spectrum

The major experimental peaks were compared with the DFT-simulated Raman frequencies. Strong correspondence was observed across the principal fingerprint regions.

Experimental (cm^{-1})	Simulated (cm^{-1})	Difference (cm^{-1})	Assignment
284.3	282.5	1.8	Ring deformation
772.7	777.8	5.1	Ring vibration
955.9	965.7	9.8	Ring breathing
1155.4	1156.9	1.5	C–O–C stretch
1249.4	1260.1	10.7	C–N / OH mode
1345.9	1347.4	1.5	CH deformation
1451.8	1431.4	20.4	CH bending
1700.7	1725.1	24.4	C=O stretch

The overall agreement between theory and experiment was excellent.

7.4 Frequency Error Analysis

Quantitative comparison between experimental and simulated peaks gave:

Parameter	Value
Matched Peaks	51
Mean Error	+0.378 cm^{-1}
RMSE	~2.1 cm^{-1}
Validation Quality	Excellent

An RMSE close to 2 cm^{-1} indicates that the selected computational method reproduced the experimental vibrational behaviour with high accuracy.

7.5 Interpretation of Minor Peak Shifts

Small differences between experimental and simulated frequencies are expected and scientifically acceptable. These shifts may arise from:

- solid-state powder effects
- intermolecular hydrogen bonding
- local sample environment
- anharmonic vibrations
- basis set limitations
- instrumental resolution

Such effects are common when comparing real measurements with gas-phase harmonic theoretical calculations.

Final Outcome of NAM Validation

The combined presence of carbohydrate peaks, ring vibrations, ether modes, CH deformations, and carbonyl bands confirms that the measured Raman spectrum is fully consistent with N-acetylmuramic acid.

The validated pure NAM spectrum can now be used as a reference for future studies involving bacterial samples, mixed biomarkers, or SERS-based diagnostics.

Final Conclusion, Scientific Significance, Limitations, and Future Scope

8. Conclusion

In the present study, a combined experimental and theoretical approach was used to establish the Raman spectral fingerprint of N-acetylmuramic acid (NAM), an important structural component of bacterial peptidoglycan. Density Functional Theory calculations were carried out using the B3LYP/6-31+G(d,p) level of theory in Gaussian 16 to predict the vibrational frequencies and Raman activities of the molecule. The raw computational outputs were post-processed through frequency scaling, normalization, and Gaussian broadening to generate a continuous simulated Raman spectrum.

Experimental Raman measurements were performed using high-purity NAM powder under multiple acquisition conditions. After systematic preprocessing involving dark subtraction, Savitzky–Golay smoothing, baseline correction, normalization, and spectral quality ranking, the optimum spectrum was selected at 25 s integration time and 80% laser power. This dataset exhibited excellent signal quality with a signal-to-noise ratio of 95.91.

The processed experimental spectrum displayed characteristic Raman bands corresponding to the carbohydrate framework, pyranose ring vibrations, ether linkages, CH deformation modes, amide-related vibrations, and carbonyl stretching. Comparison with the simulated spectrum showed strong agreement across the principal fingerprint regions. A total of 51 peaks were matched, with a mean error close to zero and an RMSE of approximately 2.1 cm^{-1}, indicating excellent correspondence between theory and experiment.

These results confirm that the measured Raman spectrum is consistent with the molecular structure of N-acetylmuramic acid. The study demonstrates that Raman spectroscopy combined with DFT modelling is a reliable strategy for vibrational assignment and molecular validation of bacterial biomarker compounds.

9. Scientific Significance of the Study

The present work provides a validated Raman reference spectrum of pure N-acetylmuramic acid. This is scientifically important because NAM is a key biomolecular component of bacterial cell walls and plays a central role in peptidoglycan architecture.

The established spectral fingerprint can support future research in the following areas:

- identification of bacterial cell wall biomarkers
- differentiation of peptidoglycan-related compounds
- development of Raman or SERS-based biosensors
- rapid pathogen screening methods
- spectral database generation for machine learning models
- sepsis and infection biomarker research

The study also demonstrates the usefulness of computational spectroscopy for interpreting complex biomolecular spectra.

10. Limitations of the Present Work

Although strong agreement was achieved, certain limitations should be acknowledged.

1. Gas-Phase Approximation in Simulation
   The DFT model represents an isolated molecule, whereas the experimental sample was measured in solid-state powder form.

2. Harmonic Approximation
   The calculated frequencies are based on harmonic vibrations and may not fully capture anharmonic effects.

3. Intermolecular Effects
   Hydrogen bonding and crystal packing interactions present in the powder sample are not explicitly included in the theoretical model.

4. Purity and Experimental Factors
   The powder purity was approximately 98%, and minor impurities or environmental factors may influence weak spectral bands.

5. Instrumental Broadening
   Experimental peak widths are affected by instrument resolution and detector characteristics.

These factors may contribute to the small frequency shifts observed between experiment and theory.

11. Future Scope

The validated NAM reference spectrum established in this study can be extended in several directions.

1. Comparative Biomarker Studies
   Raman spectra of other bacterial biomarkers such as lipoteichoic acid (LTA), lipopolysaccharide (LPS), and peptidoglycan fragments can be studied and compared.

2. Complex Sample Analysis
   The reference spectrum can be used to interpret Raman data from bacterial lysates, mixed biomolecular samples, or clinical specimens.

3. Surface-Enhanced Raman Spectroscopy (SERS)
   NAM detection sensitivity can be improved using nanostructured metallic substrates.

4. Machine Learning-Based Classification
   Peak features from NAM and related compounds can be used to train classification models for automated biomarker identification.

5. Advanced Computational Models
   Future studies may include solvent effects, periodic crystal calculations, or higher-level basis sets for improved theoretical accuracy.

6. Biomedical Applications
   The developed methodology may contribute to rapid bacterial detection platforms and point-of-care diagnostic tools.

Final Statement

This work establishes a robust foundation for the spectroscopic identification of N-acetylmuramic acid and demonstrates the potential of integrating Raman spectroscopy with theoretical modelling for biomarker-driven diagnostic research.

4.4 Computational Simulation Settings Used for Lipoteichoic Acid (LTA) Fragment Analysis

1.1 Overview of the Simulation Strategy A complete quantum mechanical simulation of intact lipoteichoic acid (LTA) is computationally challenging because LTA is a large and structurally heterogeneous polymer consisting of repeating glycerol-phosphate units with variable substitutions such as D-alanine and carbohydrate moieties. To overcome this limitation, a fragment-based computational approach was adopted. Representative molecular units corresponding to the major structural motifs of LTA were selected and individually simulated. The resulting spectra were later combined to generate a theoretical approximation of the Raman fingerprint of LTA. The three representative components selected for simulation were:

1. Glycerol-Phosphate — representing the phosphate-rich polymeric backbone
2. D-Alanine — representing amino acid substitutions attached to the backbone
3. N-Acetyl-D-Glucosamine (GlcNAc) — representing carbohydrate-associated structural contributions

This strategy allows interpretation of the major Raman-active regions of LTA while maintaining feasible computational cost.

1.2 Software and Computational Tools The following software tools were used during the molecular modeling and Raman simulation workflow:

Software	**Purpose**
PubChem	Retrieval of molecular structures and chemical identifiers
Avogadro	Molecular visualization, geometry refinement, and coordinate generation
Gaussian 16	Density Functional Theory (DFT) optimization and Raman frequency calculations
Python / Jupyter Notebook	Spectral processing, normalization, fitting, and plotting
Microsoft Excel	Storage and review of exported spectral data

1.3 Molecular Structure Preparation The molecular structures of each fragment were obtained from publicly available chemical databases. The downloaded molecular files were imported into Avogadro for structural inspection and geometry refinement. The following preparation steps were performed:

1. Import the molecular structure into Avogadro
2. Verify atomic connectivity and valency
3. Add hydrogen atoms where required
4. Perform preliminary geometry optimization using molecular mechanics
5. Export optimized Cartesian coordinates for Gaussian input generation

This preprocessing step ensured that chemically reasonable starting geometries were used for quantum calculations.

1.4 Quantum Chemical Method All Raman simulations were carried out using Density Functional Theory (DFT). Geometry optimization and vibrational frequency calculations were performed prior to Raman intensity extraction. The following computational method was employed:

Parameter	**Setting**
Electronic Structure Method	DFT
Functional	B3LYP
Basis Set	6-31G(d)
Job Type	Geometry Optimization + Frequency + Raman
Solvation Model	SMD
Solvent	Water
Symmetry	NoSymm
Integration Grid	UltraFine
SCF Convergence	Tight
Maximum SCF Cycles	512

1.5 Gaussian Input Route Section The route section used for simulation was: # opt freq=raman b3lyp/6-31g(d) scrf=(smd,solvent=water) nosymm int=ultrafine scf=(tight,maxcycle=512)

1.6 Purpose of Each Parameter

Keyword	**Purpose**
opt	Optimizes molecular geometry to minimum energy structure

freq=raman	Computes vibrational frequencies and Raman activities
b3lyp	Hybrid functional widely used for vibrational simulations
6-31G(d)	Split-valence basis set with polarization functions
scrf=(smd,solvent=water)	Includes solvent effects using implicit water model
nosymm	Prevents artificial symmetry constraints
int=ultrafine	Improves numerical integration accuracy
scf=tight	Ensures stricter electronic convergence
maxcycle=512	Allows sufficient SCF iterations for convergence

1.7 Spectral Output Processing After completion of the Gaussian calculations, Raman frequencies and activities were extracted from the output files. The simulated spectra were then processed using Python. The following post-processing steps were used:

1. Export Raman shift and intensity values
2. Apply frequency scaling factor to correct harmonic overestimation
3. Restrict spectral range to the fingerprint region
4. Normalize intensities using Min–Max normalization
5. Plot individual component spectra
6. Prepare data for composite LTA spectrum generation

A scaling factor in the range of 0.96–0.97 was evaluated to improve agreement between theoretical and experimental peak positions.

1.8 Scientific Rationale The selected computational workflow provides a practical balance between accuracy and computational feasibility. Although the model does not represent the full intact LTA polymer, it captures the dominant vibrational contributions of its principal structural units. This enables mechanistic interpretation of phosphate, amino acid, and carbohydrate Raman bands relevant to Gram-positive bacterial cell wall analysis.

Simulation of Glycerol-Phosphate as the Backbone Model of Lipoteichoic Acid

2.1 Role of Glycerol-Phosphate in Lipoteichoic Acid Lipoteichoic acid (LTA) in Gram-positive bacteria is primarily composed of repeating glycerol-phosphate or ribitol-phosphate units linked through phosphodiester bonds. These phosphate-rich repeating units form the main polymeric backbone of the molecule. Because the backbone is responsible for the most characteristic phosphate-related Raman vibrations, glycerol-phosphate was selected as the representative structural fragment for simulation. The use of glycerol-phosphate enables the study of vibrational modes associated with:

1. P–O stretching
2. P–O–C ester linkages
3. Glycerol skeletal vibrations
4. Hydroxyl-associated deformations
5. Carbon–oxygen backbone modes

These features are highly relevant for identifying LTA signatures in Raman spectra.

2.2 Molecular Source and Structure Preparation The molecular structure of glycerol-phosphate was obtained from a standard chemical database and imported into Avogadro. The following steps were followed:

1. Download the molecular structure file
2. Import into Avogadro
3. Inspect atom types and bonding pattern
4. Add missing hydrogen atoms if required
5. Perform preliminary molecular mechanics optimization
6. Save the final Cartesian coordinates for Gaussian simulation

This ensured that the starting structure was chemically stable before quantum optimization.

2.3 Gaussian Simulation Conditions

Parameter	Setting
Functional	B3LYP
Basis Set	6-31G(d)
Job Type	Geometry Optimization + Frequency + Raman
Solvation	SMD Water
Grid	UltraFine
SCF	Tight
Symmetry	NoSymm

2.4 Output Data Generated After successful convergence, Gaussian produced:

1. Optimized molecular geometry
2. Vibrational frequencies (cm^{-1})

3. Raman activities
4. Thermochemical information
5. Molecular normal modes

The Raman frequencies and activities were extracted and converted into spectral plots using Python.

2.5 Spectral Processing Performed The exported simulation data was processed as follows:

1. Raman shift values imported into Python
2. Frequency scaling factor applied (0.96–0.97 tested)
3. Fingerprint region selected (500–2000 cm^{-1})
4. Intensities normalized using Min–Max scaling
5. Publication-quality Raman plot generated

No baseline correction was applied because the spectrum originated from theoretical simulation and contained no fluorescence background.

2.6 Major Raman Regions Observed for Glycerol-Phosphate

Region (cm^{-1})	Tentative Assignment	Structural Significance
700–900	P–O–C vibrations	Phosphodiester linkage
900–1100	P–O stretching / C–O modes	Phosphate backbone
1170–1300	C–O / O–H deformation	Glycerol hydroxyl environment
1360–1490	CH_2 bending / skeletal modes	Organic backbone vibrations

A strong feature near the 1000–1100 cm^{-1} region is particularly important because phosphate stretching bands are widely reported as diagnostic signals for teichoic acid and LTA-related structures.

2.7 Importance in Composite LTA Modeling The glycerol-phosphate spectrum was later used as the backbone contributor during composite LTA reconstruction. In weighted fitting models, this fragment represents the phosphate polymer chain and contributes significantly to regions associated with phosphate vibrations. Mathematically: I_LTA = w1(I_GP) + w2(I_DAla) + w3(I_GlcNAc) Where:

1. I_GP = glycerol-phosphate spectrum
2. w1 = contribution weight of the backbone component

2.8 Scientific Interpretation The simulation confirms that glycerol-phosphate is a suitable minimal model for capturing the dominant phosphate vibrations expected from lipoteichoic acid. Since phosphate groups are central to the biochemical identity of LTA, this fragment provides the foundational spectral signature required for interpreting Gram-positive bacterial Raman spectra.

Simulation of D-Alanine as the Amino Substituent Model of Lipoteichoic Acid

3.1 Role of D-Alanine in Lipoteichoic Acid In many Gram-positive bacterial species, lipoteichoic acid contains ester-linked D-alanine substituents attached to the glycerol-phosphate backbone. These substitutions influence the surface charge, metal ion interactions, autolysin regulation, and host immune recognition of the bacterial cell wall. From a spectroscopic perspective, D-alanine introduces characteristic amino acid vibrations that are not captured by the phosphate backbone alone. D-alanine was therefore selected as the representative amino substituent model to account for Raman bands associated with:

1. NH_3^+ deformations
2. COO^- vibrations
3. C–N stretching
4. CH_3 bending modes
5. Carbon skeleton vibrations of the amino acid moiety

This fragment is important for reproducing amino-containing regions of the LTA spectrum.

3.2 Molecular Source and Structure Preparation The molecular structure of D-alanine was obtained from a standard chemical database and prepared using Avogadro. The following procedure was followed:

1. Download molecular structure data
2. Import into Avogadro
3. Inspect stereochemistry and bonding
4. Add hydrogen atoms where required
5. Perform preliminary geometry refinement
6. Export Cartesian coordinates for Gaussian calculations

3.3 Zwitterionic Consideration Free amino acids commonly exist in zwitterionic form rather than as neutral NH_2–CH(R)–COOH structures, particularly in condensed phases and aqueous environments. Therefore, interpretation of D-alanine Raman bands was carried out considering the protonated amine (NH_3^+) and deprotonated carboxylate (COO^-) forms. This is important because several major Raman peaks arise from:

1. NH_3^+ bending modes
2. COO^- stretching / rocking modes

rather than neutral amino and carboxylic acid groups.

3.4 Gaussian Simulation Conditions

Parameter	Setting
Functional	B3LYP

Basis Set	6-31G(d)
Job Type	Optimization + Frequency + Raman
Solvation	SMD Water
Symmetry	NoSymm
Grid	UltraFine
SCF	Tight

3.5 Output Data Generated After successful completion of the calculation, the following outputs were obtained:

1. Optimized geometry
2. Vibrational frequencies
3. Raman activities
4. Thermodynamic information
5. Normal mode displacements

3.6 Spectral Processing Performed The D-alanine simulation data were processed using Python:

1. Import Raman shift and intensity values
2. Apply frequency scaling factor (0.96–0.97 tested)
3. Restrict range to 500–2000 cm^{-1}
4. Normalize intensity using Min–Max scaling
5. Plot the processed Raman spectrum

3.7 Major Raman Regions Observed for D-Alanine

Region (cm^{-1})	**Tentative Assignment**	**Structural Significance**
500–650	Skeletal deformation / COO^- rocking	Amino acid backbone
800–1000	C–C stretching	Carbon skeleton
1150–1250	C–N stretching	Amino substituent linkage
1280–1380	CH_3 symmetric deformation	Methyl group

1440–1465	CH_3 asymmetric deformation	Side-chain marker
1485–1550	NH_3^+ symmetric deformation	Protonated amine
1590–1670	NH_3^+ asymmetric deformation / coupled modes	Zwitterionic form

3.8 Importance in Composite LTA Modeling The D-alanine spectrum was used as the amino substituent contributor during theoretical LTA reconstruction. It explains Raman regions that cannot be described solely by phosphate or carbohydrate fragments. Mathematically: I_LTA = w1(I_GP) + w2(I_DAla) + w3(I_GlcNAc) Where:

1. I_DAla = D-alanine spectrum
2. w2 = contribution weight of the amino substituent component

3.9 Scientific Interpretation The simulation confirms that D-alanine provides essential amino acid and zwitterionic vibrational signatures relevant to substituted lipoteichoic acid structures. Inclusion of this fragment improves interpretation of Raman bands associated with amine, carboxylate, and methyl-containing regions observed in Gram-positive bacterial spectra.

Simulation of N-Acetyl-D-Glucosamine (GlcNAc) as the Carbohydrate Model of Lipoteichoic Acid

4.1 Role of GlcNAc in Cell Wall–Associated Spectral Contributions N-Acetyl-D-Glucosamine (GlcNAc) is a biologically important amino sugar commonly found in bacterial cell wall glycans and peptidoglycan-associated structures. Although the exact composition of lipoteichoic acid varies among species, carbohydrate substituents and glycan-associated spectral features can contribute to the overall Raman response of Gram-positive bacterial surfaces. GlcNAc was therefore selected as a representative carbohydrate model to account for sugar-related and acetyl-associated vibrational modes in the composite spectrum. This fragment was used to represent Raman features associated with:

1. C–O–C vibrations
2. C–O stretching modes
3. Ring breathing vibrations
4. Amide I and Amide III bands from the N-acetyl group
5. CH / CH_2 / CH_3 deformation modes
6. Hydroxyl-associated vibrations

4.2 Molecular Source and Structure Preparation The molecular structure of GlcNAc was obtained from a standard chemical database and processed in Avogadro prior to quantum calculations. The following steps were followed:

1. Download molecular structure file
2. Import into Avogadro
3. Inspect ring geometry and bonding
4. Add hydrogen atoms where required

5. Perform preliminary geometry optimization
6. Export Cartesian coordinates for Gaussian input generation

4.3 Gaussian Simulation Conditions

Parameter	Setting
Functional	B3LYP
Basis Set	6-31G(d)
Job Type	Optimization + Frequency + Raman
Solvation	SMD Water
Symmetry	NoSymm
Grid	UltraFine
SCF	Tight

4.4 Output Data Generated Following successful convergence, the calculation provided:

1. Optimized geometry
2. Vibrational frequencies
3. Raman activities
4. Thermochemical outputs
5. Normal mode descriptions

4.5 Spectral Processing Performed The simulated GlcNAc spectrum was processed using Python:

1. Import Raman shift and intensity values
2. Apply frequency scaling factor (0.96–0.97 tested)
3. Restrict spectral window to 500–2000 cm^{-1}
4. Normalize intensities using Min–Max scaling
5. Generate publication-quality plots

4.6 Major Raman Regions Observed for GlcNAc

Region (cm^{-1})	Tentative Assignment	Structural Significance
900–1100	C–O–C / C–O stretch / ring modes	Sugar backbone

1080–1160	C–O stretching	Carbohydrate framework
1250–1310	Amide III (C–N stretch + N–H bend)	N-acetyl group
1320–1440	CH / OH bending + CH_3 deformation	Sugar ring + acetyl methyl
1440–1480	CH_2 / CH deformation	Carbohydrate skeleton
1630–1680	Amide I (C=O stretching)	Acetyl carbonyl group

4.7 Importance in Composite LTA Modeling The GlcNAc spectrum was used as the carbohydrate contributor in the fragment-based LTA reconstruction. It helps explain sugar-related and acetyl-associated bands that are not fully represented by glycerol-phosphate or D-alanine. Mathematically: I_LTA = w1(I_GP) + w2(I_DAla) + w3(I_GlcNAc) Where:

1. I_GlcNAc = GlcNAc spectrum
2. w3 = contribution weight of the carbohydrate component

4.8 Scientific Interpretation The simulation confirms that GlcNAc provides strong carbohydrate fingerprint features together with diagnostic acetyl-group Raman bands. Inclusion of this fragment improves interpretation of the mixed glycan and amide regions commonly observed in bacterial Raman spectra and supports a broader cell wall–associated model for LTA-related spectral analysis.

Generation of the Composite Lipoteichoic Acid (LTA) Spectrum Using Weighted Combination of Simulated Fragments

5.1 Rationale for Composite Spectrum Generation Lipoteichoic acid is a complex macromolecular assembly rather than a single small molecule. Its Raman spectrum arises from the combined vibrational contributions of multiple structural motifs, including the phosphate backbone, amino substituents, and carbohydrate-associated groups. Since direct quantum simulation of the complete polymer is computationally demanding, a composite spectral model was generated by mathematically combining the individually simulated spectra of the representative fragments. The three simulated contributors were:

1. Glycerol-Phosphate — backbone model
2. D-Alanine — amino substituent model
3. N-Acetyl-D-Glucosamine (GlcNAc) — carbohydrate model

5.2 Preprocessing Prior to Combination Before combining the three spectra, all datasets were standardized to ensure fair comparison. The following steps were applied:

1. Import spectral data from exported files
2. Apply frequency scaling factor to simulated Raman shifts
3. Restrict all spectra to the common analysis range (fingerprint region)
4. Interpolate onto a shared Raman shift grid

5. Normalize intensities using Min–Max normalization

5.3 Mathematical Model The composite LTA spectrum was defined as the weighted sum of the three processed component spectra. Where: I_theoretical = w1(I_GP) + w2(I_DAla) + w3(I_GlcNAc)

1. I_theoretical = composite theoretical spectrum
2. w1, w2, w3 = non-negative contribution weights

5.4 Equal-Weight Initial Model As an unbiased starting point, an equal-weight composite spectrum was generated (w1=w2=w3=1/3).

5.5 Optimized Weighting Using NNLS To improve agreement with the experimental Raman spectrum, optimized weights were obtained using Non-Negative Least Squares (NNLS). This method minimizes the difference between the composite model and the measured experimental spectrum.

5.6 Example Optimized Contribution Trend

Component	Example Weight (%)
Glycerol-Phosphate	20–25
D-Alanine	34–40
GlcNAc	40–43

5.7 Plots Generated Visual outputs included individual component spectra, composite models, and experimental overlays with residuals.

5.8 Scientific Meaning of the Composite Model The combined spectrum estimates how major structural fragments collectively contribute to the observed Raman signal.

5.9 Importance to the Thesis This composite approach demonstrates how molecular simulations can be integrated with Raman spectroscopy to interpret complex biomolecular systems.

Experimental Raman Measurement of the LTA-Related Sample and Data Processing Workflow

6.1 Objective of Experimental Measurement The experimental Raman study was performed to obtain a real vibrational spectrum from the prepared sample and compare it with the fragment-based theoretical LTA model.

6.2 Sample Material and Preparation The Raman measurement was performed on a powder sample spread uniformly onto a measurement substrate.

6.3 Instrument Conditions

Parameter	Value
Spectral Technique	Raman Spectroscopy
Laser Power	70%

Integration Time	10 s

6.4 Raw Experimental Data Obtained The exported dataset contained Raman Shift (cm^{-1}) and Measured Intensity (a.u.) columns.

6.5 Experimental Data Preprocessing Workflow Steps included data cleaning, window restriction, ALS baseline correction, and Min-Max normalization.

6.6 Why Baseline Correction Was Applied Only to Experimental Data Theoretical spectra lack the instrumental/matrix-related background (fluorescence) found in real measurements.

6.7 Output Plots Generated from Experimental Data Normalized experimental spectra were overlaid with theoretical models for interpretation.

6.8 Scientific Interpretation of Experimental Data Agreement with the model supports the presence of chemically meaningful signatures in the measured sample.

6.9 Importance to the Thesis transformation of raw Raman data into analyzable spectra is central to the study's connection between measurement and interpretation.

Comparison of Experimental Spectrum with Simulated Composite Model, Fitting Metrics, and Final Interpretation

7.1 Objective of the Comparison This step evaluated how effectively the molecular fragments reproduce the measured Raman fingerprint.

7.2 Overlay Comparison of Spectra overlays allowed visual examination of peak position and intensity trends.

7.3 Quantitative Fitting Metrics

Metric	Meaning	Desired Trend
RMSE	Average fitting error	Lower
R^2	Fraction of variance explained	Higher
Pearson Correlation	Similarity of spectral shape	Higher

7.4 Representative Results Obtained RMSE ~0.19 and Pearson Correlation ~0.34 were observed in a representative fit.

7.5 Interpretation of the Metrics The model captures broad trends, though simplified fragments don't explain all clinical complexity.

7.6 Residual Analysis Residual plots highlight missing contributors or intensity mismatches.

7.7 Region-Based Chemical Interpretation

Region (cm^{-1})	Dominant Chemical Meaning

700–900	Phosphate ester / P–O–C modes
900–1100	Carbohydrate + phosphate stretching
1200–1300	Amide III / C–N modes
1400–1500	CH_3 / CH_2 deformation
1500–1650	NH_3^+ / COO^- / Amide I

7.8 Scientific Conclusion of the Comparison The model captures characteristic regions associated with LTA functionality, supporting fragment-based spectral decomposition.

7.9 Limitations of the Current Model Simplifications include harmonic frequency scaling and a limited number of fragments.

7.10 Future Scope Includes peak broadening and adding more cell wall components.

7.11 Final Thesis Statement This study established a framework for interpreting LTA-related Raman spectra, providing a route toward future diagnostic technologies.

4.4 Work Completed on Lipopolysaccharide (LPS)

Introduction and Selection of Target Molecule

Lipopolysaccharide (LPS) was selected as the target molecule for this study because it is a major structural component of the outer membrane of Gram-negative bacteria and plays a critical role in endotoxin-mediated inflammatory responses, including sepsis. Due to its biomedical significance, identifying and understanding the Raman fingerprint of LPS is important for future diagnostic applications.

LPS is a structurally complex glycolipid composed of three major regions: lipid A, the core oligosaccharide, and the O-antigen polysaccharide chain. Because of its high molecular size, structural heterogeneity, and multiple flexible linkages, direct quantum chemical simulation of the complete LPS molecule is computationally demanding. Therefore, a fragment-based modelling strategy was adopted.

In this approach, smaller representative molecular units corresponding to key structural regions of LPS were selected and individually simulated. These fragments were later combined mathematically to reconstruct and interpret the experimental Raman spectrum.

Structural Components Considered

The following representative components were selected:

Selected Fragment	Structural Role
3-Deoxy-D-manno-octulosonic acid (KDO)	Core oligosaccharide marker
L-glycero-D-manno-heptose	Core sugar region
D-Glucosamine	Lipid A backbone
Myristic Acid	Fatty acid chain
Phosphoric Acid	Phosphate functionality

These fragments were chosen because they represent the major chemical domains expected to contribute to the vibrational fingerprint of the full molecule.

Collection of Molecular Structures and Preparation for Simulation

After selecting the representative fragments, the next stage involved obtaining their molecular structures and preparing them for quantum chemical simulation. Accurate molecular geometry is essential because Raman frequencies and intensities are highly dependent on the optimized atomic arrangement of the molecule.

Source of Molecular Structures

The selected fragments were obtained from publicly available chemical databases such as PubChem in three-dimensional structure format. Each molecule was downloaded individually using its standard chemical identity.

Molecules Prepared for Simulation

Molecule	Purpose in Study
3-Deoxy-D-manno-octulosonic acid (KDO)	Core oligosaccharide contribution
L-glycero-D-manno-heptose	Sugar-region contribution
D-Glucosamine	Backbone sugar contribution
Myristic Acid	Lipid chain contribution
Phosphoric Acid	Phosphate group contribution

Geometry Preparation

The downloaded structures were inspected and refined before simulation. Initial molecular cleanup and geometry preparation were carried out to remove unrealistic bond arrangements and ensure chemically valid starting coordinates.

The preparation process included:

1. Verification of atom types and bonding pattern
2. Addition of missing hydrogen atoms (if required)
3. Correction of valency or structural inconsistencies
4. Preliminary geometry cleanup
5. Saving structures in a Gaussian-compatible format

Pre-Optimization Using Molecular Mechanics

Before running Density Functional Theory (DFT) calculations, each structure was pre-optimized using molecular mechanics methods. This helps reduce steric strain and provides a stable starting structure for quantum calculations.

The following sequence was used:

- Steepest Descent Optimization: 200 steps
- Conjugate Gradient Optimization: 500 steps

This two-stage minimization improved convergence behavior during Gaussian calculations.

Quantum Chemical Simulation of Raman Spectra

After molecular preparation, each fragment was subjected to quantum chemical calculations to generate theoretical Raman spectra. The purpose of this stage was to obtain vibrational frequencies and Raman activities for every selected molecular unit so that their individual spectral contributions could be analyzed.

Software Used

Gaussian software was used for all simulations.

Computational Method

Density Functional Theory (DFT) was selected because it provides a suitable balance between computational accuracy and practical runtime for medium-sized molecules.

Level of Theory

- Functional: **B3LYP**
- Basis Set: **6-31G(d)**

This combination is widely used for vibrational spectroscopy studies and gives reliable molecular frequencies for organic and biomolecular fragments.

Type of Calculations Performed

For each molecule, the following calculations were carried out:

1. **Geometry Optimization**
   To locate the minimum-energy stable structure.
2. **Frequency Calculation**
   To compute vibrational modes and confirm that the optimized structure corresponds to a real minimum.
3. **Raman Activity Calculation**
   To obtain Raman intensities associated with each vibrational mode.

General Gaussian Input Settings

The following computational settings were used:

- opt : geometry optimization
- freq=raman : vibrational frequency + Raman activity calculation
- scf=tight : stricter self-consistent field convergence
- integral=ultrafine : improved numerical integration grid
- Multi-core processing enabled
- High memory allocation for stable execution

Example Route Section

opt freq=raman b3lyp/6-31g(d) scf=tight integral=ultrafine

Molecules Simulated Individually

Separate calculations were completed for:

- 3-Deoxy-D-manno-octulosonic acid (KDO)
- L-glycero-D-manno-heptose
- D-Glucosamine
- Myristic Acid
- Phosphoric Acid

Each molecule was treated independently to obtain its own Raman fingerprint.

Output Obtained from Gaussian

For every fragment, the following outputs were extracted:

1. Vibrational frequencies (cm^{-1})
2. Raman activities / relative intensities
3. Optimized molecular geometry
4. Log file for validation
5. Checkpoint file for saved wavefunction data

Validation of Optimized Structures

The frequency calculations were also used to confirm that no significant imaginary frequencies were present, indicating that the optimized geometry corresponded to a stable molecular minimum.

Extraction, Conversion, and Processing of Simulated Spectral Data

After completion of the Gaussian calculations, the raw simulation outputs were converted into usable spectral datasets for comparison with the measured Raman spectrum. Gaussian provides vibrational frequencies and Raman activities in text format; therefore, additional processing was required to transform these values into plot-ready spectra.

Data Extracted from Simulation Files

From each Gaussian output file, the following parameters were collected:

1. Raman shift values (cm^{-1})
2. Raman activity / intensity values
3. Peak positions
4. Relative peak strengths

These values formed the basis of the simulated spectra.

Conversion to Structured Data Files

The extracted spectral data for each fragment was organized into spreadsheet format for easier handling and reproducibility.

Separate Excel files were created for:

- 3-Deoxy-D-manno-octulosonic acid
- L-glycero-D-manno-heptose
- D-Glucosamine
- Myristic Acid
- Phosphoric Acid

Each file contained two primary columns:

Column	Description
Raman Shift	Frequency in cm^{-1}
Intensity	Relative Raman response

Frequency Scaling

Theoretical harmonic frequencies generally differ slightly from experimental values. To reduce this deviation, an empirical scaling factor was applied to the simulated frequencies.

Purpose of scaling:

- Correct harmonic overestimation
- Improve alignment with measured peaks
- Enhance practical comparability

Normalization of Intensities

Because each molecule produced different absolute intensity ranges, all simulated spectra were normalized before comparison.

Min–max normalization was used:

$$I_{norm} = \frac{I - I_{min}}{I_{max} - I_{min}}$$

This converted all spectra to a common intensity scale between 0 and 1.

Interpolation to Common Raman Grid

Different spectra may contain peak values at different Raman shift points. To enable direct point-by-point comparison, every simulated dataset was interpolated to a common x-axis grid.

Common Grid Used

- Raman range: **200–2000 cm^{-1}**
- Grid spacing: **1 cm^{-1}**

Why This Was Necessary

Interpolation ensured:

1. Equal data length for all spectra
2. Fair mathematical comparison
3. Compatibility with fitting algorithms
4. Uniform plotting and overlay analysis

No Baseline Correction Applied to Simulated Data

Baseline correction was intentionally not applied to simulated spectra because theoretical Raman outputs do not contain fluorescence background or detector drift.

Experimental Raman Measurement and Optimization of Acquisition Parameters

After preparing the simulated spectra, experimental Raman measurements were carried out to obtain the real vibrational fingerprint of the powder sample. The purpose of this step was to generate a high-quality experimental spectrum that could be directly compared with the computational model.

Sample Used

Commercially available lyophilized powder sample of lipopolysaccharide obtained from *Escherichia coli* O111:B4 was used for the study.

Physical Form

- Powder sample
- Dry-state measurement
- Direct Raman acquisition without chemical labeling

Experimental Objective

1. Record Raman spectra of the powder sample.
2. Evaluate the effect of acquisition settings on signal quality.
3. Identify the best condition for further analysis.
4. Use the optimized dataset for comparison with simulation.

Why Optimization Was Required

The quality of Raman spectra strongly depends on measurement conditions. Poor settings may result in:

- weak peaks
- excessive noise
- unstable baseline
- overheating of sample
- poor reproducibility

Therefore, multiple parameter combinations were tested systematically.

Parameters Varied During Trials

The following acquisition parameters were examined:

1. **Integration Time**
   Duration of signal collection for each scan.
2. **Laser Power**
   Relative excitation intensity used during measurement.
3. **Accumulations**
   Number of repeated scans averaged together.

Trial Measurements

Several combinations of settings were tested, including shorter and longer acquisition times. Representative conditions included:

- 10 s integration, 20 power, 5 accumulations
- 20 s integration, 20 power, 5 accumulations
- Additional intermediate trials
- 60 s integration, 20 power, 30 accumulations

Selection Criteria for Best Spectrum

The final condition was selected based on:

- stronger Raman signal
- clearer peak visibility
- reduced noise
- stable baseline behavior
- repeatability across scans
- no visible sample damage

Final Optimized Experimental Setting

The condition that produced the most reliable spectrum was:

- **Integration Time:** 60 s
- **Laser Power:** 20%
- **Accumulations:** 30

Reason for Choosing 60 s / 20% / 30

- Longer integration increased photon collection and signal intensity.
- Moderate laser power reduced risk of thermal damage.
- Multiple accumulations improved averaging and signal-to-noise ratio.

Final Experimental Dataset

The optimized spectrum was saved and used as the primary experimental reference dataset for subsequent analysis.

Example file naming format:

sec-60_power-20_i-30

Preprocessing of Experimental Raman Data

The optimized experimental spectrum obtained from the powder measurement was further processed before comparison with the simulated spectra. Raw Raman data often contains baseline drift, instrumental noise, and intensity variations that can affect quantitative fitting. Therefore, preprocessing was performed to generate a clean and comparable dataset.

Input Data

The selected experimental file corresponding to the optimized condition was imported from spreadsheet format into the analysis environment.

Example dataset:

sec-60_power-20_i-30

Initial Data Handling

The raw file was inspected and cleaned using the following steps:

1. Removal of blank rows
2. Removal of non-numeric entries
3. Conversion of columns into numeric format
4. Sorting according to Raman shift
5. Duplicate x-value handling (if present)

Raman Shift Window Selection

Only the relevant fingerprint region was retained for final analysis:

- **200–2000 cm^{-1}**

This range was selected because it contains the most informative vibrational bands associated with carbohydrates, phosphate groups, and hydrocarbon structures.

Baseline Correction

Experimental spectra can contain a sloping or curved background caused by fluorescence or instrumental drift. To remove this unwanted background, baseline correction was applied.

Method Used

Asymmetric Least Squares (ALS) baseline correction.

Purpose of Baseline Correction

- Remove fluorescence contribution
- Correct baseline slope
- Improve peak visibility
- Enable accurate fitting

Smoothing

To reduce random noise while preserving peak shape, smoothing was applied.

Method Used

Savitzky-Golay filter

Purpose of Smoothing

- Improve signal-to-noise ratio
- Retain peak positions
- Improve visual clarity

Intensity Normalization

The corrected spectrum was normalized to place intensity values on a common scale.

Min–max normalization was used:

$$I_{norm} = \frac{I - I_{min}}{I_{max} - I_{min}}$$

Why Normalization Was Necessary

- Remove scale dependence
- Enable fair comparison with simulations
- Improve fitting stability

Final Processed Experimental Spectrum

After preprocessing, the experimental dataset consisted of:

- clean baseline
- reduced noise
- normalized intensity scale
- selected Raman range
- ready-to-fit format

Composite Spectrum Construction and Mathematical Fitting

After preparing both simulated fragment spectra and the processed experimental spectrum, the next stage was to mathematically combine the simulated spectra and compare them with the measured data. The aim of this step was to determine how strongly each fragment contributed to the final observed spectrum.

Why Composite Modelling Was Needed

The experimental Raman spectrum arises from multiple chemical domains present in the target material. Since each simulated fragment represents only one structural region, a combined spectrum was required to approximate the complete experimental fingerprint.

Therefore, the individual fragment spectra were integrated using two modelling approaches:

1. Equal-weight composite model

2. Optimized weighted model using Non-Negative Least Squares (NNLS)

7.1 Common Comparison Grid

Before fitting, all spectra were aligned on the same Raman shift axis.

Grid Used

- Range: **200–2000 cm⁻¹**
- Resolution: **1 cm⁻¹**

This ensured direct point-by-point comparison between all datasets.

7.2 Equal-Weight Composite Model

As an initial unbiased reference model, all simulated spectra were averaged equally.

Formula

$$I_{composite} = \frac{I_1 + I_2 + I_3 + I_4 + I_5}{5}$$

Where:

- I_1= KDO spectrum
- I_2= Heptose spectrum
- I_3= D-Glucosamine spectrum
- I_4= Myristic Acid spectrum
- I_5= Phosphoric Acid spectrum

Purpose

- Baseline comparison model
- No prior bias toward any fragment
- Simple first approximation of the measured spectrum

7.3 Optimized Weighted Model (NNLS)

To obtain a more realistic reconstruction, a weighted fitting approach was used.

Method Used

Non-Negative Least Squares (NNLS)

Mathematical Model

$$I_{fit} = w_1 I_1 + w_2 I_2 + w_3 I_3 + w_4 I_4 + w_5 I_5$$

Where:

- w_1, w_2, w_3, w_4, w_5 are fitted weights
- All weights are constrained to be **non-negative**

Weight Normalization

After fitting, weights were normalized such that:

$$\sum w_i = 1$$

Why NNLS Was Selected

NNLS was chosen because:

1. Negative spectral contributions are not physically meaningful
2. It minimizes fitting error efficiently
3. It provides interpretable relative contributions
4. It is widely used in mixture and spectral decomposition studies

7.4 Meaning of Fitted Weights

The fitted weights do **not** represent exact chemical concentration or mass fraction. Instead, they represent the relative spectral contribution of each fragment required to reconstruct the measured Raman spectrum.

A larger weight indicates that the fragment spectrum better explains important features of the experimental data.

Model Evaluation, Peak Comparison, and Interpretation of Results

After generating the composite spectra, the next stage involved quantitative evaluation of model performance and interpretation of the fitted results. The optimized model was compared with the experimental spectrum to determine how effectively the simulated fragments reproduced the measured Raman fingerprint.

8.1 Performance Metrics Used

To assess the quality of fitting, multiple statistical metrics were calculated.

Root Mean Square Error (RMSE)

RMSE measures the average difference between the experimental and fitted spectra.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Where:

- y_i= experimental intensity
- $\hat{y}_i$= fitted intensity
- n= number of data points

Lower RMSE indicates better agreement.

Coefficient of Determination (R^2)

R^2 measures how much variance in the experimental spectrum is explained by the fitted model.

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Higher R^2 indicates stronger explanatory performance.

Correlation Coefficient

This metric measures similarity in spectral shape between fitted and experimental curves.

Values closer to 1 indicate stronger similarity.

8.2 Final Quantitative Results

Equal-Weight Model

Metric	Value
RMSE	0.18853
R^2	0.14877
Correlation	0.41652

NNLS Optimized Model

Metric	Value
RMSE	0.18505
R^2	0.17995
Correlation	0.50784

Interpretation

The optimized model outperformed the equal-weight model by:

- lower fitting error
- higher explained variance
- stronger shape similarity

This confirmed that a weighted combination of fragments is more realistic than equal averaging.

8.3 Final Optimized Weights

Fragment	Weight	Percentage
KDO	0.473	47.3%
Heptose	0.258	25.8%
D-Glucosamine	0.153	15.3%
Myristic Acid	0.117	11.7%
Phosphoric Acid	0.000	0.0%

8.4 Scientific Interpretation of Weights

KDO

The highest weight was assigned to KDO, indicating that sugar-rich core-region vibrations strongly influenced the measured spectrum.

Heptose

The second highest contribution further supported the importance of core oligosaccharide components.

D-Glucosamine

Moderate contribution suggested participation of the backbone region.

Myristic Acid

A measurable but smaller contribution indicated the presence of hydrocarbon chain vibrations.

Phosphoric Acid

Zero fitted weight does not imply absence of phosphate functionality. It indicates that this isolated reference did not add unique explanatory power beyond overlapping features already captured by other fragments.

8.5 Peak Region Comparison

The optimized model showed useful agreement with several important Raman regions:

Region (cm^{-1})	Interpretation
~900–1100	Sugar / ring / backbone modes
~1130	Hydrocarbon chain vibrations
~1300–1450	CH deformation modes
~1600 region	Higher-energy structural modes

8.6 Residual Analysis

Residuals were calculated as:

$$Residual = Experimental - Fitted$$

Residual plots were used to identify regions where mismatch remained.

These deviations are expected because real powdered biomaterials exhibit:

- intermolecular interactions
- hydrogen bonding
- packing effects
- conformational disorder
- instrumental broadening
- environmental effects

Such behavior is not fully captured by isolated fragment simulations.

Final Conclusion, Significance of the Study, and Future Scope

This study established a complete workflow for interpreting a complex Raman spectrum through the integration of experimental spectroscopy, quantum chemical simulation, and mathematical fitting. A fragment-based strategy was successfully used to overcome the difficulty of modelling a large biomolecular system directly.

9.1 Overall Conclusion

Representative molecular fragments were selected to describe the major chemical domains of the target structure. Each fragment was simulated independently using Density Functional Theory to generate theoretical Raman spectra. Experimental Raman measurements were then performed under multiple acquisition conditions, and the optimum dataset was selected based on signal quality, peak clarity, and reproducibility.

After preprocessing and normalization, the simulated spectra were combined mathematically using two approaches: equal-weight averaging and Non-Negative Least Squares (NNLS) optimization. The optimized model provided better agreement with the experimental spectrum than the equal-weight model.

The final fitting results demonstrated that carbohydrate-rich fragments were the dominant contributors to the measured Raman fingerprint, while hydrocarbon chain contributions were also present. This confirms that fragment-based modelling can provide chemically meaningful interpretation of complex spectra.

9.2 Scientific Significance

The significance of this work can be summarized as follows:

1. Demonstrated a reproducible workflow combining experiment and simulation.
2. Showed that large molecular systems can be interpreted using smaller representative units.
3. Applied quantitative fitting instead of subjective visual comparison.
4. Generated interpretable weight values representing relative spectral contribution.
5. Created a scalable framework for future biomolecular Raman studies.

9.3 Key Technical Achievements

- Successful simulation of five representative molecular fragments
- Optimization of Raman acquisition parameters
- Development of a preprocessing pipeline for raw spectra
- Construction of composite spectral models
- Implementation of NNLS fitting
- Statistical evaluation using RMSE, R^2, and correlation
- Identification of dominant contributors through fitted weights

9.4 Limitations of the Present Study

Although the workflow produced meaningful results, certain limitations remain:

1. Isolated fragments do not fully capture intermolecular interactions.
2. Environmental effects such as hydration and packing were not explicitly modelled.
3. Harmonic frequency calculations may not reproduce all peak shifts exactly.
4. Only a limited set of representative fragments was used.
5. Experimental spectra may contain broadening and instrument-dependent variations.

These limitations are common in first-order modelling studies and do not reduce the value of the developed methodology.

9.5 Future Scope

The present work can be extended in several directions:

1. Include additional or linked structural fragments.
2. Use advanced DFT methods or solvent-corrected models.
3. Perform conformer averaging for flexible molecules.
4. Apply region-wise fitting for better local agreement.
5. Integrate Surface-Enhanced Raman Spectroscopy (SERS) studies.
6. Use machine learning for classification and prediction.
7. Extend the workflow to real biological matrices such as serum or plasma.
8. Develop rapid biosensing platforms based on the identified spectral fingerprints.

9.6 Final Statement

The study demonstrates that computational fragment modelling combined with Raman spectroscopy and constrained spectral fitting is an effective and scientifically robust strategy for analyzing complex molecular fingerprints. This approach provides both practical interpretability and a strong foundation for future translational diagnostic research.

4.5 COMPARATIVE ANALYSIS OF BIOMARKERS

To evaluate the feasibility of Raman/SERS-based sepsis biomarker detection, three important bacterial biomarkers—N-acetylmuramic acid (NAM), lipoteichoic acid (LTA), and lipopolysaccharide (LPS)—were investigated using computational modelling and experimental Raman spectroscopy. These biomarkers were selected because they represent key structural components of bacterial cell walls and are clinically relevant indicators of Gram-positive and Gram-negative infections. A comparative analysis was performed to assess their spectral behavior, theoretical–experimental agreement, modelling approach, and diagnostic potential.

Biomarker	Type	Method Used	Key Result
NAM	Gram + / PGN marker	Full molecule DFT + Raman	Strong validation
LTA	Gram + marker	Fragment model + Raman	Good composite fit
LPS	Gram – marker	Fragment model + Raman	Quantitative fitting

Among the three biomarkers, NAM demonstrated the strongest agreement between simulated and experimental Raman spectra, with multiple characteristic peaks successfully validated. This indicates that NAM is a robust and chemically interpretable target for bacterial biomarker sensing. LTA, due to its larger and more complex polymeric structure, required a fragment-based modelling strategy using representative subunits such as glycerol-phosphate, D-alanine, and GlcNAc. The reconstructed composite spectrum showed meaningful correspondence with experimental observations, validating the usefulness of the fragment approach. LPS, a major Gram-negative endotoxin marker, also required fragment-based simulation and demonstrated effective quantitative fitting, highlighting its relevance for Gram-negative sepsis detection.

Overall, the comparative study confirms that all three biomarkers exhibit diagnostically significant Raman signatures and are promising candidates for future SERS-based rapid sepsis diagnostic platforms.

5. CONCLUSION

Sepsis remains a major global healthcare challenge due to its rapid progression, high mortality rate, and dependence on timely diagnosis. Conventional diagnostic approaches such as blood culture, immunoassays, and molecular techniques often require prolonged processing time, specialized laboratory infrastructure, and trained personnel, which can delay critical clinical intervention. The present project addressed this challenge by exploring a Raman spectroscopy–based analytical framework for rapid identification of sepsis-related bacterial biomarkers using Surface-Enhanced Raman Spectroscopy (SERS)-oriented sensing concepts. The work focused on three important bacterial biomarker classes: N-acetylmuramic acid (NAM), lipoteichoic acid (LTA), and lipopolysaccharide (LPS), representing key structural signatures of Gram-positive and Gram-negative pathogens.

The primary achievement of this study was the successful development of an integrated computational–experimental workflow for biomarker analysis. Density Functional Theory (DFT)-based Raman simulations were carried out using Gaussian software to predict vibrational fingerprints of the selected biomarkers. For structurally complex molecules such as LTA and LPS, fragment-based modelling strategies were designed to overcome computational limitations while retaining chemical interpretability. Experimental Raman spectra of biomarker powders were acquired under optimized acquisition conditions, followed by preprocessing using baseline correction, smoothing, normalization, and spectral quality evaluation. Quantitative fitting approaches, including Non-Negative Least Squares (NNLS), were employed to compare theoretical and experimental spectra.

The results demonstrated strong agreement between simulated and measured spectra, particularly for NAM, where multiple characteristic peaks were successfully validated with low frequency error. Fragment-based reconstruction of LTA and LPS spectra further showed that representative molecular units could explain key biochemical regions associated with phosphate groups, carbohydrate structures, amino substituents, and hydrocarbon chains. These findings confirm that Raman spectroscopy, supported by computational modelling, can provide chemically meaningful signatures for bacterial biomarker detection.

The practical significance of this project lies in its potential application toward rapid, label-free, and portable diagnostic systems for early sepsis screening. A future SERS-based biosensor built upon these validated biomarker fingerprints could support intensive care units, emergency departments, neonatal care, and resource-limited settings by enabling faster decision-making and earlier treatment initiation. Beyond sepsis, the developed methodology may also be extended to pathogen identification, infection monitoring, and point-of-care molecular diagnostics.

Certain limitations remain in the present work. The study primarily used purified powder standards rather than complex biological samples such as serum or plasma. Full macromolecular structures were approximated through fragment models, and harmonic DFT calculations required empirical scaling. In addition, clinical validation and real-time sensor integration were beyond the current project scope.

Future work should focus on fabrication of optimized plasmonic SERS substrates, detection of biomarkers in biological fluids, machine learning–based spectral classification, multiplex biomarker sensing, and validation using clinical samples. With further development, the proposed approach has strong potential to evolve into a translational diagnostic platform for rapid sepsis detection and broader infectious disease management.

6. NOVELTY

6.1 Unique Idea / Innovation

The project titled Sepsis Biomarkers Analysis Using SERS-Based Sensor introduces a novel computational–experimental framework for rapid detection of sepsis-associated biomarkers using Raman spectroscopy and Surface-Enhanced Raman Spectroscopy (SERS) principles. The core innovation lies in combining plasmonic nanostructure-based sensing, theoretical Raman modelling, and multi-biomarker spectral validation into a unified diagnostic strategy. Instead of relying solely on conventional biochemical assays, the proposed work uses intrinsic molecular vibrational fingerprints of biomarkers such as N-acetylmuramic acid (NAM), lipoteichoic acid (LTA), and lipopolysaccharide (LPS) for label-free detection.

A second important innovation is the use of fragment-based DFT spectral reconstruction for large and complex bacterial biomolecules. This approach enables theoretical interpretation of macromolecular Raman spectra that are otherwise computationally expensive to model as full structures. The integration of simulation, experimental spectroscopy, and quantitative spectral fitting creates a scalable pathway for future intelligent biosensor development.

6.2 Difference from Existing Solutions

Current sepsis diagnostic methods such as blood culture, ELISA, chemiluminescence assays, and PCR-based systems primarily depend on centralized laboratories, reagents, and multi-step workflows. These methods often require several hours to days for actionable results. Existing biosensors may focus on single biomarkers and may not provide molecular-level spectral interpretation.

In contrast, the proposed work is based on direct optical fingerprinting of biomarkers using Raman/SERS signatures. It eliminates the need for complex labeling chemistry and supports rapid analysis with minimal sample preparation. Unlike many published studies that report only experimental spectra, this project establishes a validated computational reference library using DFT simulations, improving confidence in biomarker peak assignments.

6.3 Technical Advantages

The proposed system offers multiple technical advantages. Raman/SERS sensing provides high molecular specificity, enabling selective identification of biomarker signatures. The use of plasmonic Ag/Au substrates can significantly enhance weak Raman signals, improving sensitivity for low-concentration detection. Computational modelling reduces ambiguity in spectral interpretation and accelerates assay development. The framework is compatible with future machine learning integration for automated classification, anomaly detection, and multiplex decision support. In addition, the method has potential for miniaturization into portable or handheld diagnostic devices.

6.4 Practical Advantages

From a translational perspective, the proposed work can support faster bedside screening, decentralized healthcare delivery, and improved accessibility in resource-limited settings. Compared with expensive laboratory analyzers, a portable SERS-based platform can reduce dependence on centralized infrastructure and trained specialists. The label-free approach may lower recurring reagent costs and simplify maintenance. Such systems can be deployed in intensive care units, neonatal care, emergency departments, ambulances, and rural clinics where rapid triage is critical.

6.5 Why It Matters

Sepsis management is highly time-sensitive, and delayed diagnosis increases mortality risk. Therefore, innovation in rapid diagnostics has direct clinical and societal impact. The proposed work matters because it moves sepsis testing toward fast, affordable, portable, and intelligent molecular diagnostics. By enabling earlier detection of infection-related biomarkers, the system has potential to improve treatment decisions, reduce hospital burden, and enhance patient outcomes. Furthermore, the developed methodology can be extended to other infectious diseases and biomarker-driven healthcare applications.

6.6 Comparative Table

Parameter	Existing Solutions	Proposed Work
Detection Principle	Culture / Immunoassay / PCR	Raman/SERS Molecular Fingerprinting
Response Time	Hours to Days	Minutes to Rapid Screening
Cost	High Instrument + Reagent Cost	Lower Long-Term Operational Cost
Portability	Limited / Lab-Based	High Potential for Portable Deployment
Biomarker Scope	Often Single or Limited Panel	Multi-Biomarker Expandable Framework
Interpretation	Standard Assay Output	Spectral + Computational Validation
Intelligence	Basic Automation	AI/ML Compatible
Accessibility	Centralized Hospitals	Wider Point-of-Care Reach
Sample Preparation	Multi-Step	Minimal / Simplified
Scalability	Moderate	High with Sensor Miniaturization

7. POTENTIAL SOCIETAL AND MARKET IMPACT OF THE PROPOSED INNOVATION

The proposed project, Sepsis Biomarkers Analysis Using SERS-Based Sensor, has significant potential to create measurable impact across healthcare, society, and the diagnostics market. By addressing the critical challenge of delayed sepsis diagnosis, the innovation can contribute to improved patient survival, reduced healthcare burden, and advancement of affordable point-of-care medical technologies.

7.1 Societal Impact

Sepsis is a time-sensitive medical emergency in which early diagnosis directly influences treatment outcomes. A rapid biomarker sensing platform can help clinicians identify high-risk patients earlier and initiate timely intervention, thereby reducing mortality and long-term complications. This is especially relevant for neonates, elderly individuals, immunocompromised patients, and critically ill populations who are highly vulnerable to severe infection.

The proposed system also supports broader societal goals of healthcare accessibility and affordability. Conventional diagnostics are often concentrated in tertiary hospitals and urban laboratories, limiting access for rural populations and resource-constrained communities. A portable SERS-based platform can enable decentralized testing in smaller hospitals, community health centers, ambulances, and remote clinics. Faster screening can reduce anxiety for patients and families while improving confidence in medical decision-making. In the long term, such technologies contribute to equitable healthcare delivery by bringing advanced diagnostics closer to underserved populations.

7.2 Healthcare and Industrial Impact

For hospitals and clinical laboratories, the proposed innovation can improve diagnostic workflows by reducing dependence on lengthy culture-based testing and reagent-intensive assays. Rapid biomarker screening may support emergency triage, ICU monitoring, neonatal intensive care, and post-surgical infection surveillance. Earlier detection can enable targeted treatment strategies, reduce empirical antibiotic overuse, and support antimicrobial stewardship programs.

The project is also relevant to the medical device and diagnostics industry. It creates opportunities for development of portable Raman analyzers, disposable SERS chips, clinical software platforms, and integrated decision-support systems. Manufacturers of nanomaterials, biosensors, optics, and healthcare instrumentation can benefit from the growing demand for rapid diagnostic tools. The innovation further aligns with trends in personalized medicine, digital diagnostics, and connected healthcare systems.

7.3 Economic Impact

Delayed sepsis diagnosis significantly increases healthcare expenditure due to prolonged ICU stays, repeated laboratory testing, advanced supportive care, and extended hospitalization. A rapid screening platform can reduce these costs by enabling earlier intervention and optimized treatment pathways. Hospitals may benefit from improved bed utilization, faster clinical decisions, and reduced burden on centralized laboratories.

From a macroeconomic perspective, reducing sepsis-related mortality and morbidity improves workforce productivity and lowers indirect costs associated with long-term disability. Lower-cost decentralized diagnostics can also reduce patient travel expenses and out-of-pocket healthcare spending, which is particularly important in emerging economies such as India.

7.4 Market Potential

The market potential for rapid sepsis diagnostics is substantial due to rising global incidence of infections, increasing ICU admissions, and growing demand for point-of-care testing. Target users include:

- Multispecialty hospitals and ICUs
- Emergency departments
- Neonatal and pediatric care units
- Diagnostic laboratories
- Rural healthcare centers
- Ambulance and mobile care services
- Defense and disaster-response healthcare units

India represents a strong growth market due to its large population, expanding healthcare infrastructure, and increasing adoption of indigenous medical technologies under initiatives such as *Make in India* and *Digital Health Mission*. Globally, the solution can address demand in both developed and low-resource healthcare systems.

7.5 Commercialization Opportunity

The proposed work has strong potential for translation into a startup or licensable healthcare product. Commercial pathways may include development of:

1. Portable Raman/SERS sepsis screening devices
2. Disposable biomarker detection cartridges or chips
3. Cloud-enabled clinical interpretation software
4. AI-based infection risk scoring platforms
5. Integrated hospital diagnostic dashboards

Partnership opportunities exist with hospitals, medtech manufacturers, nanotechnology companies, and public health agencies. Technology transfer, patent filing, and incubator-based startup support can accelerate commercialization.

7.6 Long-Term Impact

In the long term, the innovation can strengthen national healthcare preparedness, improve infectious disease management, and support the transition toward smart diagnostics. It aligns with broader goals of preventive healthcare, digital transformation, and affordable indigenous technology development. Beyond sepsis, the same platform can be adapted for other infectious diseases, inflammatory disorders, and biomarker-based screening applications. Therefore, the proposed project has the potential to evolve from an academic research initiative into a scalable healthcare solution with lasting societal and economic value.

8. CHALLENGES OR RISK FACTORS ASSOCIATED WITH THE PROJECT

The project titled **Sepsis Biomarkers Analysis Using SERS-Based Sensor** offers strong translational potential; however, like any emerging medical technology, it is associated with technical, experimental, operational, financial, regulatory, and market-related challenges. Identifying these risks at an early stage is important for successful development, validation, and deployment.

8.1 Technical Challenges

One of the primary technical challenges is achieving consistent and reproducible SERS enhancement. Raman signal intensity strongly depends on the nanostructure geometry, surface roughness, particle distribution, and hotspot formation on the sensing substrate. Small fabrication variations can lead to signal fluctuations between batches. In addition, weak biomarker concentrations in biological samples may produce low signal-to-noise ratios, requiring advanced preprocessing and optimized optics.

Another challenge is spectral interference from complex matrices such as blood, plasma, or serum. Biological fluids contain proteins, salts, lipids, and metabolites that may mask target biomarker signatures. Instrument calibration, baseline drift, laser stability, and detector sensitivity are also important factors affecting data quality. If machine learning is integrated in future versions, risks may include overfitting, poor generalization, and dataset bias.

8.2 Experimental and Validation Risks

At the current stage, much of the work has been performed using purified standards and controlled laboratory conditions. Translating these results to clinical samples introduces uncertainty due to patient variability, biomarker heterogeneity, and differing disease stages. Limited access to annotated sepsis samples can slow validation studies. Reproducibility across instruments, operators, and environmental conditions must also be established before real-world deployment.

Another risk is that some biomarkers may exist at concentrations below the practical detection limit of the initial prototype. Multiplex detection of several biomarkers simultaneously may also require additional optimization of substrate chemistry and spectral deconvolution methods.

8.3 Operational Challenges

For clinical adoption, the device must integrate smoothly into hospital workflows. If operation requires highly trained personnel, adoption may be limited. User interface complexity, maintenance requirements, calibration frequency, and consumable replacement can affect usability. Rural or resource-limited deployment may face additional challenges such as unstable power supply, limited technical support, and environmental dust or humidity.

8.4 Financial Risks

Prototype fabrication involving nanomaterials, Raman optics, detectors, and precision instrumentation can be expensive during early development. Scaling from laboratory prototypes to manufacturable products requires additional investment in tooling, quality assurance, and supply chains. Funding constraints may delay iterative testing, clinical validation, or regulatory submission.

8.5 Regulatory and Ethical Challenges

As a healthcare diagnostic technology, the system would require compliance with relevant medical device regulations, safety standards, and performance validation protocols before commercialization. Depending on deployment geography, approvals may involve CDSCO (India), CE marking, or FDA pathways. If cloud analytics or AI models are used, data privacy, cybersecurity, and secure patient data handling become important considerations. Clinical claims must be supported by robust evidence to avoid misuse or overreliance.

8.6 Market and Adoption Risks

Healthcare systems may be cautious in adopting new diagnostics without strong clinical evidence, cost-benefit justification, and interoperability with existing systems. Competition from established laboratory assays and rapidly evolving point-of-care technologies may create pricing pressure. Procurement cycles in hospitals can also be slow, especially for new vendors or startups.

8.7 Critical Success Factors

The long-term success of the project depends on achieving high analytical sensitivity, specificity, reproducibility, affordability, ease of use, and portability. Additional critical factors include clinician trust, stakeholder partnerships, manufacturing readiness, and scalability for large-scale deployment.

8.8 Mitigation Strategies

These risks can be reduced through phased development and rigorous engineering practices. Standardized substrate fabrication protocols, robust calibration methods, and replicate testing can improve reproducibility. Validation with progressively complex samples—from standards to spiked samples to real clinical samples—can strengthen confidence. Collaborations with hospitals, clinicians, and industry partners can support data access and translational testing. Cost can be reduced through design-for-manufacturing strategies and modular hardware selection. Early regulatory planning, documentation, and quality management systems can accelerate future approvals. Finally, user-centered design, intuitive software interfaces, and operator training can improve adoption in real healthcare environments.

8.9 Risk Summary Table

Risk Category	Challenge	Impact	Mitigation
Technical	Variable SERS enhancement	Reduced accuracy	Standardized substrate fabrication
Experimental	Limited clinical samples	Delayed validation	Hospital collaborations
Operational	Training and maintenance needs	Lower usability	Simple UI and training modules
Financial	High prototype cost	Slower development	Grants, partnerships, phased scaling
Regulatory	Approval requirements	Delayed commercialization	Early compliance planning
Market	Resistance to new technology	Slow adoption	Clinical evidence and cost-benefit proof

In summary, while the project faces realistic challenges, these risks are manageable through structured development, interdisciplinary collaboration, and evidence-driven validation.

9. ANY PATENT OR EQUIVALENT FILED

At present, **no patent or equivalent intellectual property (IP) has been formally filed** for the project titled **Sepsis Biomarkers Analysis Using SERS-Based Sensor**. The current stage of the work has primarily focused on problem definition, literature review, computational modelling, Raman spectral validation of target biomarkers, and proof-of-concept experimental studies. These activities have been essential for establishing scientific feasibility and identifying potential areas of innovation before initiating formal IP protection.

A preliminary novelty and patent landscape assessment has been conducted to understand the existing state of technology in sepsis diagnostics, biosensors, Raman spectroscopy platforms, and SERS-based detection systems. This review indicates that while several patents exist in the areas of immunoassays, molecular diagnostics, substrate fabrication, and portable Raman instrumentation, opportunities remain for innovation in integrated, low-cost, multi-biomarker, and point-of-care sepsis sensing solutions.

Based on the current findings, the project contains multiple components that may be considered patentable after further technical validation and prototype maturation. These may include:

- Novel Ag/Au nanostructured SERS substrate designs optimized for sepsis biomarker detection
- Multi-biomarker spectral detection workflow for rapid sepsis screening
- AI-assisted Raman spectral classification and diagnostic decision-support system
- Portable point-of-care sensing device architecture for bedside use
- Low-volume sample handling cartridge integrated with SERS detection platform

The future IP strategy would involve completing prototype development, performing repeatability and clinical validation studies, generating comparative performance data, and documenting clear claims of novelty over existing technologies. Following these milestones, a **provisional patent application** may be considered as the first step toward formal protection, followed by complete specification filing, technology transfer, licensing, or startup commercialization depending on translational outcomes.

In summary, no IP filing has been made to date; however, the project demonstrates sufficient innovation potential for future patent consideration after further development and validation.

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